

Digital Twin–Enabled Networks for Real-Time Network Optimization

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Abstract—This survey gives a detailed description of Digital Twin Networks (DTNs) as a new technology of the real-time optimization of a contemporary 5G and 6G networks when growing complexity and a heterogeneous traffic are making the old method of managing the network less efficient. DTNs generate virtual versions of physical network infrastructures at high fidelity, and allow them to be continually observed, analyzed, and safely subjected to what-if testing in an environment that is not interfered with by the production traffic. The survey examines major architectural features of DTNs, such as division into physical, cyber and service space to facilitate the two-way data synchronization. It has also outlined key enabling technologies, including Graph Neural Networks (GNNs) to predict Quality of Service (QoS) accurately and Deep Reinforcement Learning (DRL) to slice and allocate resources in an adaptive manner. Besides, practical use and case studies such as IGP weight optimization and frameworks like DTRAN to optimize the end-to-end Radio Access Network are discussed in the paper, which show considerable network performance improvements including a lower latency and a better allocation of resources. Lastly, the survey discusses the key research challenges, such as real-time synchronization, large-scale infrastructure scalability, AI model generalization, data management, and security, and presents future research directions, which involve technologies, such as Generative AI and blockchain, to promote the creation of fully autonomous, self-optimizing network infrastructures.

I. INTRODUCTION

THE evolution of modern next-generation communication network technologies including fifth generation and emerging sixth generation systems, Software Defined Networking, Network Function Virtualization has resulted in transformative shift in global telecommunication landscape. This modernization impacts to networks in many aspects such as rapid growth of connected devices, high data traffic and diverse service requirements. Fifth generation and future sixth generation networks are expected to support ultra-reliable low-latency communication(uRLLC), massive machine-type communications(mMTC) and enhanced mobile broadband(eMBB). These all efforts ultimately increase the complexity of networks due to introduction of highly dynamic and heterogeneous traffic environment characteristic. Management and optimization of higher complex networks become challenging for network operators with traditional solutions. Traditional network management techniques rely on static configurations, offline analysis and manual optimization processes. These approaches are not suitable to handle rapid changes in traffic patterns and service demands in modern

networks. Modern network systems become growing need for intelligent and adaptive mechanisms to enable continuous monitoring, analyzing, and optimizing network performance in real time. To address these challenges in modern communication networks, several software-based solutions have been introduced to enhance efficient management, monitoring and optimization while allowing intelligent network automation. Among these solutions, the Network Digital Twin(NDT) has emerged as a key innovation which enables real time network processes.

A Digital Twin(DT) is a virtual replica of physical network system that continuously updates with the real-time data from the actual environment. In the context of communication networks, DT technology enables the creation of a virtual model of the network infrastructure having bidirectional data synchronization with its real-world counterpart. DT allows continuous analysis, testing and optimization of the network with support of the digital replicas. Optimization algorithms can be simulated and validated in the digital replica before interacting with the physical network. According to the results shown in the DT environment, operations can be performed on the physical network ensuring more robust operation, optimization and automation. This proactive approach prevents adverse impacts on network performance and establishes a robust foundation for autonomous operations. The integration of artificial intelligence (AI) and machine learning (ML) within the Network Digital Twin transforms networks by enabling intelligent forecasting of resource behavior, automating real-time decision-making, and significantly improving Quality of Service (QoS), energy efficiency, and fault tolerance. NDT provides an advanced sandbox environment to training and validating advanced AI agents before they are deployed in critical infrastructure.

This survey aims to provide a comprehensive review of the current research landscape regarding digital twin-enabled networks for real-time optimization. The lack of intelligence and compatibility with data-driven operations of legacy networks has created new research gaps for future network areas. This paper analyzes architectural advancements, main enabling technologies, diverse application domains and the inherent challenges associated with modern digital twin networks.

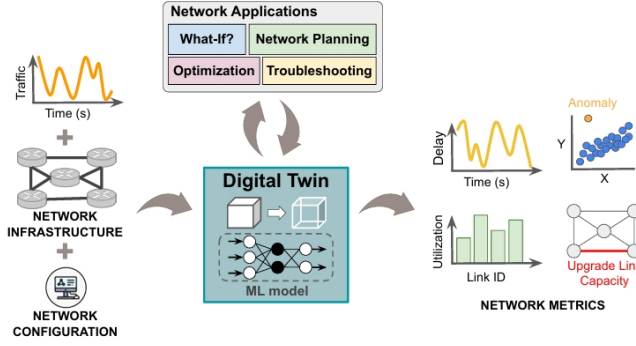


Fig. 1. Architecture of Digital Twin Network [1]

II. ARCHITECTURE OF DIGITAL TWIN ENABLED NETWORKS

NDT is a data-driven concept that leverages network data to build a digital representation of network. The data is gathered from sources like real-world networks, network testbeds and network simulation tools. The gathered large amount of data is used for build DTN model with the support of advanced Deep Learning(DL) technologies. DL technologies can perform efficiently at the complex network scenarios.

A. Digital Twin Network Model

Fig 1 represents a general reference architecture of the NDT. The central component of the architecture is the DT, which implements a network model that replicates the physical network. This model takes network state description(e.g., traffic, topology, routing,scheduling policies) as the input and generates some network-related metrics or features(e.g.,utilization, delay, anomalies, packet drops) as its outputs [1]. The input and outputs are problem and context dependent. Network operators can test any input values, even if the values cause network disruptions. This is because NDT is operated in a safe environment isolated from real-world network [1].

B. Network Optimizer

A *network optimizer* is integrated with DT Networks(DTN) for enhance the operational efficiency of the networks. Fig 2 shows the process of DTN with the optimizer. Network operator sends their requirement to the optimizer(step 1). Optimizer search the best network configuration that fulfills the requirement specified by network operator(e.g.,change the maximum link utilization). Performance matrices of the DTN are evaluated and optimizer continuously search until the stopping condition is met(step 3). The best solution found so far is applied to the real-world network(step 4). The network configurations can be evaluated using simulation/emulation tools like Mininet before applied to the physical infrastructure. This optimization process can be closed loop and human interaction is not necessary [1]. Network optimizer must respond to real-time variations in the network like changes in traffic, topology and resource usage to achieve efficient dynamic network management. But the unexpected link failures

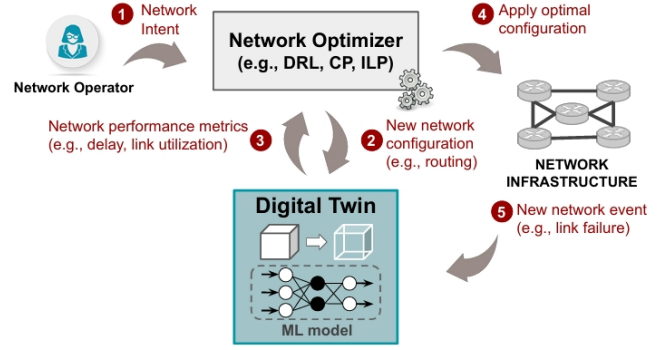


Fig. 2. Architecture of Network Digital Twin for real-time network optimization.

and varying traffic patterns results in unpredictable network resource usage surges. Deep Reinforcement Learning(DRL) is a novel technology which has shown the capability of better network functioning in dynamic circumstances and complex scenarios [2], [3]. But DRL shows often limitations of less-than-ideal configurations that creates optimization issues. Also, wide solution spaces makes it difficult for DRL to identify best solutions in the cases like optimum resource allocation. Therefore, it is needed to explore new methods to increase the optimization performance and several works began to merge DRL with conventional optimization techniques such as Integer Linear Programming (ILP) [4]. A new type of optimization algorithm was introduced based on the Software Defined Networking(SDN) which promoting centralized view of networks [5]. DTN with integrated optimizer can solve network planning, congestion minimization, load balancing and Traffic Engineering problems in dynamic networks.

C. Training The Digital Twin

Modeling a NDT is involve with collecting large relevant dataset. There are two issues related to this as the speed of model building and model training. The major challenge under the model training part is converting the collected data into input for which the model can work. Datasets can be obtained from real-world systems, simulation tools, and from different non production testbeds. According to the previous researches [6], [7] proved that, collecting data from production environments is sometimes not feasible. Therefore, current researches focuses on producing training datasets in non-production scenarios. Maintaining the data accuracy for further training is another main challenge related to training NDT model with large state space. Generalization [8] is needed to be perform on these states for achieve the ability of the model to make accurate decisions based on limited subsets of data from past experiences. Researchers typically partition data into training dataset, validation dataset, and test dataset [8] sets to ensure the NDT truly understands network behavior without memorizing specific training examples called overfitting. This dataset separation allows the NDT to be evaluated on its ability to handle unpredictable traffic scenarios and topology changes that were not part of its initial training. In real-world network deployments, network environments change rapidly and

models are required to be retrained at every new event. Widely used ML strategies like dropout and regularization are used to prevent overfitting, but it can sometimes introduce bias or limit the model's accuracy [9]. The proposed efficient alternative is the combination of Reinforcement Learning (RL) and generalization techniques. This approach enables the NDT to handle incrementally acquired data by balancing exploitation (use past experiences) [9] with exploration (able to make decisions with new experiences) [9]. This balance makes sure NDT remains both accurate and adaptable. Hence, NDT can provide reliable optimizations even with the unpredictability of live network infrastructures.

D. Real-Time Synchronization

Real-time synchronization is the main difference between a Network Digital Twin (NDT) and a traditional network simulator. A simulator usually looks at how a network behaves under certain conditions and gives results for a certain situation. In contrast, an NDT continuously updates itself by collecting real-time data from the physical network, such as traffic patterns, latency information, and device status. This constant syncing makes sure that the digital twin accurately shows how the real network is doing right now. This kind of real-time visibility is very important for traffic engineering, which is when network operators try to manage traffic flows in a way that reduces congestion and packet loss and makes better use of resources by load balancing. Several technological advances have helped NDT grow. Cloud-based infrastructures and technologies for processing large amounts of data make it possible to analyze the huge amounts of network data that people all over the world create every day. Software-Defined Networking and Network Function Virtualization are two examples of technologies that give networks the flexibility and programmability they need to change their settings on the fly. These technologies let the digital twin system interact to the physical network and help with real-time network optimization.

III. ENABLING TECHNOLOGIES FOR DIGITAL TWIN BASED NETWORK OPTIMIZATION

A wide range of cutting-edge technologies make digital twin-based network optimization possible. These technologies make it possible for the system to keep the physical and virtual replicas in sync in real time. This makes sure that the digital twin shows the current state of the network. In addition, advanced analytical techniques support intelligent decision-making within digital twin environments.

A. Deep neural networks

NDTs are referred to as a new generation of network modeling tools that leverage ML techniques to build an accurate data-driven digital network representation [10]. To train these network models, we can use data from real-world networks, dedicated network testbeds, or network simulation tools. This data should be diverse enough to cover a wide representation of potential scenarios that the network operator wants to mimic



Fig. 3. Inputs and outputs of RouteNet-F [11].

(e.g., various congestion levels, link failures). In this context, recent deep learning (DL) techniques are of interest as they enable building accurate digital models of complex network environments.

Recent DL models offer several advantages with respect to traditional network modeling tools (e.g., simulators, queueing theory). For example, DL-based models have shown state-of-the-art performance when modeling fixed networks [10], outperforming well-known analytical models based on queueing theory. In addition, they are easy to parallelize and have a low execution cost compared to traditional network simulation tools (e.g., OMNet++, ns-3).

Graph neural networks (GNNs) are a DL-based architecture recently proposed by the ML community to model relational information. GNNs capture graph dependencies using a message-passing algorithm between the graph's entities (nodes and edges). Since communication networks are fundamentally represented as graphs, GNNs offer unique advantages for network modeling when compared to traditional NN architectures (e.g., multilayer perceptron, recurrent NN). In recent years, GNNs have demonstrated outstanding performance in solving a wide variety of network-related problems. In this context, GNNs may be a central technology to enable the construction of ML-based network models that can generalize to different network topologies, configurations, and traffic distributions.

RouteNet-Fermi, also known as RouteNet-F [11], is a GNN model of communication networks proposed by the same authors as TwinNet, Ferriol-Galmés et al. [12]. It uses the topology of the network (including link bandwidths, queueing policy of each interface, and the number of queues at each interface and their respective sizes), the traffic flows between each source-destination node (described by the average bandwidth demand, and the packet size and time distributions), and the route each traffic flow takes as input. The resulting output of RouteNet-F is the predicted delay, loss, and jitter of each traffic flow. Their results show that RouteNet-F can generalize and accurately predict QoS metrics on unseen topologies and traffic flows. It serves as an ML-based network model that is capable of predicting QoS metrics much faster than traditional tools.

B. Deep Reinforcement Learning (DRL)

Deep Reinforcement Learning (DRL) is a powerful decision-making tool that learns the best network setups by interacting with a virtual environment all the time. When conditions change, traditional optimization methods like ILP or CP have to start over from scratch. DRL algorithms, on the

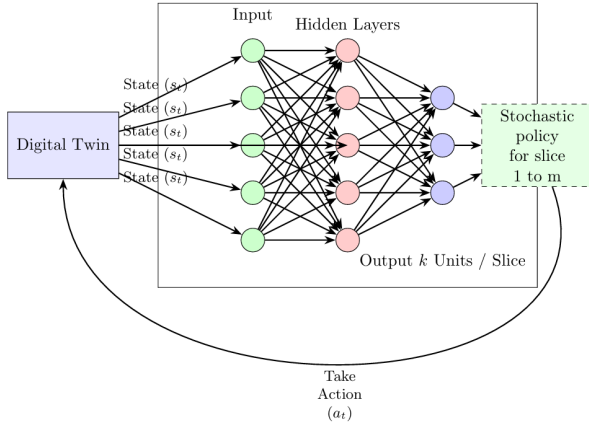


Fig. 4. DRL agent's fully connected neural network [13].

other hand, use information from past optimizations to adjust to the dynamic nature of modern 5G and 6G networks [1].

Deep Deterministic Policy Gradient (DDPG) is a reinforcement learning algorithm that works well for managing resources in networks because it can work with continuous action spaces. DDPG can choose exact values, which lets it make smooth and flexible changes to resources like bandwidth allocation. This is different from traditional reinforcement learning methods, which only let it choose from a limited number of options. This feature makes it useful for slicing 5G and 6G networks, where resources like Resource Blocks (RBs) need to be dynamically assigned to different services, such as enhanced Mobile Broadband (eMBB). For example, when drone-based base stations are moving, DDPG can constantly change how resources are allocated as the distance between users changes. This can cut down on latency by a lot. The algorithm uses an actor-critic structure, where the actor suggests a resource setup and the critic uses feedback from the Digital Twin network model to judge how well it works. Fig 4 The architecture which depicts how the DT provides state to each input node of the DRL agent's fully connected neural network

Popular reinforcement learning algorithm for making stable and effective decisions in complicated network tasks like routing optimization and managing virtual network functions (VNFs) is Proximal Policy Optimization (PPO). It makes learning more stable by limiting how much a new policy can change from the old one. This stops learning from becoming unstable or collapsing. PPO has been used in Optical Transport Networks (OTN) to find the best routes for traffic and to manage VNF migration by keeping an eye on IoT server conditions to balance power use and latency. But as the network gets bigger, PPO can take a lot of processing power, so researchers often use it with Evolutionary Strategies (ES) to make training faster and easier to scale. New research shows that Evolutionary Strategies (ES) can train an AI much faster than PPO methods [14], but only if you have enough computers and the network isn't too big. As networks get bigger, the "guessing game" of mutations slows things down and makes training take a lot longer [14].

To find the best policy, Deep Reinforcement Learning (DRL) algorithms need to try and fail thousands of times. Doing such tests directly on a real network would be very dangerous and could cause service outages all the time. The Digital Twin is a safe "sandbox" where you can train and test DRL algorithms:

- **Exploration without Risk:** The DRL agent can test thousands of network configurations in the digital twin without affecting real production traffic.
- **Real-Time Synchronization:** The twin is continuously updated with real-world metrics such as signal strength, user mobility, and traffic bursts every few milliseconds, ensuring that the DRL agent learns from up-to-date network conditions.
- **Proactive Optimization:** Once the DRL agent finds the best setup in the digital twin, that same setup can be safely used on the real network as a tested, high-performance solution..

C. Network Infrastructure and Management

Network Infrastructure and Management Virtualization and softwareization provide the necessary framework to deploy and manage the digital twin life cycle.

Software defined Network serves as the essential backbone of Network Digital Twins by separating the control plane from the data plane, which is the most important part of Network Digital Twins. This lets a centralized controller see the whole network from one place. The digital twin can keep the virtual copy of the network in sync with the real network by continuously collecting real-time topology and traffic data. The SDN controller acts as the execution engine when the twin's AI models, which are often powered by Deep Reinforcement Learning, find the best configurations. It sends those decisions down to physical hardware through its southbound interfaces in a closed-loop automation cycle [15]. This tight integration makes it possible to do proactive "what-if" scenario testing, advanced features like Network Slicing and VNF, and quick responses to changing conditions in places like vehicular networks and IoT [16]. This means that SDN is both the "eyes" that see the network's true state and the "hands" that put intelligent optimizations back into the real world

Network Function Virtualization (NFV) is the best infrastructure for Digital Twin Networks because it replaces dedicated physical hardware with software-based Virtualized Network Functions (VNFs) that run on generic servers. This makes network elements inherently flexible and digital by nature. The ETSI framework has three main parts [16],

- 1) **NFV Infrastructure (NFVI)**, which provides the virtualized computing foundation.
- 2) **Virtual Network Functions (VNFs)**, which are software copies of real physical devices.
- 3) **Management and Orchestration (MANO) layer**, which controls their entire life-cycle from creation to termination.

NFV gives digital twins a dynamic, programmable "body" to live in, instead of just being static models. Operators can

automatically deploy virtual copies of whole network topologies using standardized templates like TOSCA and YANG. Network Slicing lets dedicated twins be assigned to individual virtual lanes for more detailed performance monitoring [15]. Because VNFs are only software-based, AI models built into the twin can predict and start live migrations to improve load balancing and energy efficiency. They can also let you test updates and configurations on virtual functions without any risk before making changes to live hardware [17]. NFV turns the digital twin from a passive observation tool into an active, self-managing, and cost-effective copy of the whole physical network.

Edge and Cloud Computing, are the foundational elements of the infrastructure that Digital Twin Networks needs to operate on both high fidelity and real time responsiveness. Cloud computing offers the scalable storage and central computing capacity to perform complex workloads such as the deep learning model training, ray tracing simulations and absorbing large volumes of heterogeneous big data [17], which is suited to long term planning and twins management of cities. Conversely, edge computing enables the process of supporting the ultra-low latency requirements of 5G and 6G applications by moving processing near data origin, providing near real time feedback to applications with time sensitive response requirements such as autonomous vehicle and augmented reality [18]. Contemporary architectures integrate the two in a cooperative Cloud-Edge-End architecture, with edge nodes responding fast to local applications and cloud nodes performing long-term storage, workload distribution, and AI acceleration with special hardware, such as GPUs and TPUs. The combination gives the computational reflexes and intelligence needed to support the Data-intensive next generation applications creating hundreds of millions of terabytes each day, as well as guarantee Quality of Service with dynamic load balancing and dedicated network slice.

D. Data Modeling and Storage

To be optimized effectively, a Digital Twin Network (DTN) needs a well-organized database that can maintain the complicated relationship between network elements. Communication networks are graphs structured by nature, so that even the traditional relational databases are not always enough to represent these relationships. Thus, network data are depicted and processed with the help of graph databases and knowledge graphs, e.g., Neo4j. [19] Graph based data modeling is used to maintain the complex associations among the various sources of data, including association among user equipment and base stations, in architecture such as DRAN. Moreover, the knowledge graphs make it possible to construct higher-level representations of the network, which can be used to generate reasoning data that can be used in advanced operations of the digital twin, including autonomous routing and detailed network performance analysis [15].

E. Technologies used to enable security and privacy

Since Digital Twin Networks (DTNs) deal with sensitive operational data, security is an important part of making sure

the system is reliable and trustworthy. To address these issues, we can point out some specific technologies,

- **Permissioned Blockchain for Trust and Data Integrity:** Permissioned blockchain offers an authenticated and decentralized registry to store digital twin information and AI model modifications. It further ensures that only the authorized users can access the network via permission control and the immutable aspect of the blockchain records assures that the data is intact and cannot be manipulated. This also creates mutual trust in a shared network environment of multiple organizations in the same network environment like in the Industrial IoT systems.
- **Federated Learning for Privacy Protection:** Federated Learning facilitates distributed learning of AI without the need to transfer raw data to a central data point. The local training of the model takes place on edge devices and it is only the learned parameters or gradients which are sent to the central system. This method will minimize the possibility of sensitive data exposure and, nevertheless, provide the global model with the advantages of the insights generated on a variety of devices.

F. Performance Modeling

NDTs uses data-driven methods with trained data models which are lightweight, efficient and reliable for addressing problems inaccuracy, inefficiency and inflexibility. A research is conducted for identify requirements for network performance evaluation and compares data-driven performance models based on data, models and applications [20]. Researches have proposed a comprehensive DTN architecture with three layers. Those layers are focusing on continuous data collection in runtime and reconstruction internal relations of collected data for represent the physical network state [20]. There are four general network scenarios explored as planning, operation, optimization and upgrade for determine specific performance model requirements [20].

- **Planning:** Designers need to ensure the overall performance of planned network meet the requirement of given topology, traffic demand and configuration.
- **Operations:** Engineers are willing to understand about real-time performance variations and quick responses under potential anomalies.
- **Network Optimization:** It involves the tuning the network for obtain maximum benefits with new configuration policies. The model provides an ideal zero-risk environment to execute optimization decisions [20].
- **Network upgrade:** Typically, it involves with topology changes and link expansions.

For the above scenarios, there are three key requirements that performance models must strictly achieve and those are as follows:

- **Fidelity:** It assesses the accuracy of performance metrics compared to the physical network across various scenarios. There are three fidelity levels as Long-term(Provides statistical results over time), Short-term(Describes de-

tailed changes over time slots) and One-to-one(Precisely models each packet's behavior in the network) [20].

- **Efficiency:** It focuses on the speed and resource consumption of performance models. Vital for real-time operation and optimization to enable frequent performance evaluations. Models need to be cost-effective and efficient in evaluating performance metrics.
- **Flexibility:** Essential for adapting to network changes that supporting planning, optimization and upgrades effectively.

Performance evaluation tasks associated with metrics such as delay, throughput, loss rate and flow completion time (FCT) [20]. Some performance models mainly focus on inference or estimation, while others concentrate on performance prediction. Performance interface or estimation models lie in evaluating current performance and performance prediction models lie in evaluating future performance. Models have wide range of applications across different tasks and scenarios and that showcases the broad potential of performance models [20]. Diversified data sources are used by models for enhance fidelity and practical applications. It evolves with machine learning techniques progress for more refined models.

Performance modeling of NDT relies on specific data sources, inputs, and outputs [20].

- **Data source**
 - **Simulation environments** are used due to cost-effectiveness and ease of deployment.
 - **Testbeds** are built by researchers for data collection, offering data closer to real-world scenarios with fast speeds.
 - **Production networks** are used for directly measure data eliminating simulation and traffic model issues.
- **Inputs**
 - **Diverse Inputs:** Studies use multiple discrete or continuous variables as inputs for different tasks. It incorporates various variables like timestamp, RTT, and response time improves the accuracy and generalization of performance models.
 - **Limited Inputs:** Using fewer features as inputs to reduce costs and improve efficiency under specific scenarios and tasks.
- **Outputs**
 - **Performance Metrics:** General outputs include metrics like RTT, delay, throughput, loss rate and FCT, typically evaluating long-term average performance under steady-state conditions.
 - **Probabilistic Metrics:** It generates metric distributions allows users to assess performance probabilistically. Some cases estimate performance distributions under "What-if" scenarios to provide percentage guarantees for SLA requirements.

Model Selection and customization in performance modeling for digital twin networks progress through three key stages:

1) Classical-Method Stage:

This stage Utilizes classical methods like Causal

Bayesian Network (CBN) and Support Vector Regression (SVR) for performance modeling. CS2P applies Hidden Markov Model (HMM) for prediction in video bitrate adaptation. These approaches offer efficiency and fidelity but lack flexibility for evolving networks.

2) Vanilla-NN Stage:

These models are based on graph structures and novel NN techniques. They offer greater flexibility for modeling long-term network performance. Neural Networks (NNs) are employed for performance modeling with abundant data availability and computing advancements.

3) Customized-NN Stage:

In this stage, Researchers focus on detailed performance modeling with models like Deep-Q, MimicNet, and xNet. Customized NN models offer accuracy, scalability, and flexibility for complex network characteristics.

The evolution from classical-method to customized-NN stages in performance modeling for digital twin networks reflects a shift towards more flexible, accurate and detailed approaches. Also it leverages the advancements in neural network techniques and data-driven methodologies.

IV. PERFORMANCE EVALUATION IN NETWORK

A. Evaluation based on RouteNet-E

There was a case study which aims to analyze in more detail the application of a state-of-the-art ML-based NDT for performance evaluation in fixed IP networks. In addition, they perform some experiments where they leverage a ML-based NDT for routing optimization in a QoS-aware optimization use case take as a reference specific AI model, called RouteNet-E [10]. RouteNet-E is a state-of-the-art Graph Neural Network (GNN based model designed to accurately predict network delays. The model takes as input a network state description consisting of the network topology, traffic matrix, and routing policy. Based on these inputs, it mimics network behavior and generates end-to-end delay predictions for all network paths. To train the model, a dataset containing 100,000 samples was generated using network topologies with 25–50 nodes, simulated with the packet-level network simulator OMNeT++. For evaluation, a separate test dataset consisting of 500 samples was created using larger topologies ranging from 50 to 300 nodes [10].

The results of the experiments with various models on network topologies can be seen in Fig 6 of the paper with a maximum of 300 nodes. The results are measured by the mean absolute percentage error between the predicted and the actual values of the network simulator delays. The findings reveal that at a network size that is comparable to the training data (25–50 nodes), the machine learning models especially RouteNet-E and RNN model have lower prediction errors than the baseline of the analytical queuing theory (QT) [10]. However, machine learning models may also be faced with challenges when tested on data that no longer satisfies the training conditions, e.g. larger network topologies or different traffic patterns. RNN model performance declines significantly with the increase in network size. Instead, RouteNet-E predicts more consistently and with better accuracy even much larger networks since its

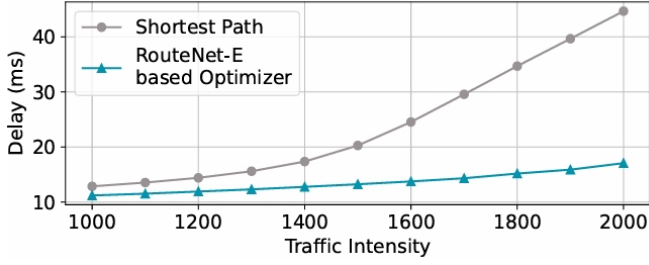


Fig. 5. Delay-aware routing optimization using a ML-based NDT (RouteNet-E) [11].

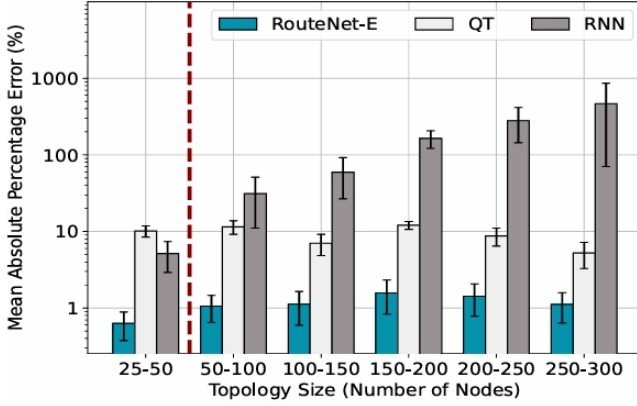


Fig. 6. Comparison of several state-of-the-art methods for network performance evaluation [11].

graph neural network architecture is able to effectively being able to capture relationships in the network and is capable of working on larger topologies.

The NDT-optimizer has been tested on a synthetically-created 25-node network topology with a wide-range of traffic rates, beginning with low load, up to the highly congested states [10]. Fig 5 showed that the optimizer was always able to reduce end to end delay than the shortest path routing which was used as the baseline, proving that optimizer was effective in the optimization of performance in the network depending on the available traffic conditions.

B. Evaluation based on performance of DRL

The idea suggested is a Digital Twin (DT) combined with a Deep Reinforcement Learning (DRL) agent that optimizes the allocation of network resources in 5G/6G NTN-based networks [13]. The physical network in this architecture constantly transmits real-time information to the digital twin which transforms the information into system states that are inputted to the DRL agent. The DRL agent is founded on the Deep Deterministic Policy Gradient (DDPG) algorithm that produces an action like bandwidth allocation of various network slices. This is modeled by the digital twin which gives the agent feedback in the form of rewards which helps the agent to keep on learning and making better decisions. Under this cyclic process, the system is able to effectively coordinate the network slice resources and deal with dynamic situations

like bursts of traffic and mobile flying base stations. The key novelty of the framework is three-fold, namely, real-time synchronization between the physical network and the digital twin, non-terrestrial network (NTN)-specific optimization, and adaptive resource allocation using DRL [13].

The suggested DT-DRL model was tested on a 5G network simulator, with 50 user equipment (UEs), with 20 MHz bandwidth, and flying base station at 10 m/s. The digital twin is updated at 10 ms and the DDPG agent will receive training of 50,000 episodes. The framework was compared to the performance of the scheme of static allocation and proportional fairness-based heuristics approach. The fig 7a shows that the DT-DRL strategy can enhance the performance of networks. It had a mean latency of 33 ms at 50 seconds, which was less than that of static allocation (54 ms) and the heuristic approach (48 ms). As shown in fig7b the resource utilization was also more with the framework of 92% as compared to the 78% in the case of the static allocation and the 85% in the case of the heuristic method [13].

As illustrated in the fig 7c, the DT-DRL approach had a smaller jitter (2.1 ms) than static allocation (8.8 ms) and heuristic method (2.9 ms). The primary factors behind these enhancements are that the digital twin can forecast the demand in the traffic and that the adaptive allocation of resources through the DRL agent adaptively responds to the changing network conditions in terms of traffic variation and movement of the flying base station.

V. DIGITAL TWIN APPLICATIONS & USE CASES

A. Troubleshooting

Network failures can be occurred due to many factors and debugging them in modern networks is complex and time consuming process. In legacy networks troubleshooting is done by human interaction with networking tools. Network operators can use DTN to replicate past network failure scenarios and analyze their impact on network performance. They can find sources that produce network failures, making it easier to find configuration errors and prevent service disruptions happen in future.

B. What-if Analysis

DTN acts as a safe sandbox for the physical network to perform what-if analysis. Different configurations are applied to the DTN for understand the potential impact on the network. What-if analysis allow operators to detect network misconfiguration, bottleneck links, disaster condition network performance observation and predict future network behaviour under different events. There is no any impact to real network while performing those operations.

C. Network Planning

The increasing size and traffic of the network creates requirement of network planning. In order to accommodate this growth in users and traffic of applications, network need periodical upgrades. ISPs increase certain link capacities or add new connections on their existing network infrastructure

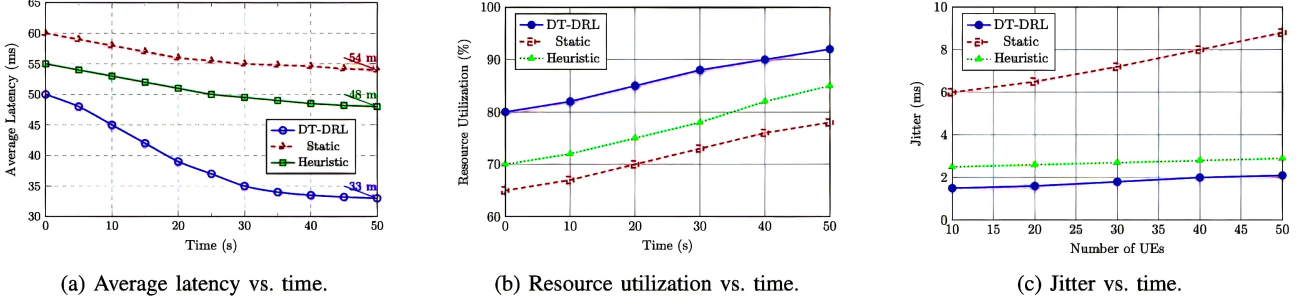


Fig. 7. Performance comparison of proposed method with baseline techniques for eMBB slice [13]

is an example for upgrades. Modern networks becoming larger and complex, thus it is difficult to existing solutions scale up to more larger networks. DTN can conduct efficient and accurate performance estimations to check network will run out of resource, based on growth in users. Also, the performance estimates provides help to plan the optimal upgrade solution that can meet the future growth.

D. Network Anomaly Detection

DTN is developed with the behaviour of real-world networks and operators have access to estimation of expected network behavior. When the real world network behaviour deviates from DTN's behaviour, it can indicate the anomaly in real-world network. Those anomalies can be occurred at different locations in a network and different data sources can be used for such anomalies.

E. Education and Training

DTN can be used as a safe playground for education and certification platform in network field. DTN can be used in cybersecurity learning for evaluate the effected of network attacks and performance measures.

Table I shows summarized applications and special use cases of NDTs.

TABLE I
APPLICATIONS OF NETWORK DIGITAL TWINS

Application	Example Use Case
Troubleshooting	Network operators can replicate past network scenarios with the NDT to identify the root cause of service disruptions.
What-If Analysis	The NDT acts as a safe sandbox where different configurations can be tested to understand their impact on network performance.
Network Planning	The NDT can help estimate when an existing network will reach capacity and require upgrades.
Anomaly Detection	When real-world network behavior deviates from normal patterns, the NDT can detect anomalies and identify potential root causes.

F. DTRAN for Radio Access Network Optimization

1) *Problem Statement*: Existing DT solutions mainly focus narrowly on modeling specific sections of networks such

as core-network [21] [22], Radio Access Network [23] and wireless networks [24]. These solutions lack the flexibility and adaptability required for fully autonomous, end-to-end network control because they are designed to optimize specific tasks and services. Furthermore, current solutions heavily rely on existing and pre-trained models, which are not much compatible with the unpredictable and highly dynamic nature of upcoming 6G networks.

2) *Solution Framework*: To address the current challenges, RAN optimization using the Digital Twin (DTRAN) framework proposes a comprehensive approach spanning from core to edge networks that enhance adaptability, accuracy, and autonomous network environment. DTRAN framework provides operators with in-depth visibility into network behavior, capturing the effects of different network components. DTRAN aligns with the long-term vision of fully autonomous networks because of real-time communication with its physical counterpart. This framework contains innovative real-time analytics and decision-making algorithms to identify and address emerging network challenges effectively.

Several quantitative metrics are used for evaluating the performance and DTRAN framework utilizes three primary metrics that mentioned below [19].

- **Twinning rate** is the frequency at which the digital model is recalculated or calibrated to remain synchronized with the physical network
- **Age of twin** is a metric that measures the freshness of the DT model. It inversely proportional to the twinning rate, since higher twinning rate reduces the age of twin [25].
- **Fidelity level** represents the accuracy and explainability of the model. High fidelity level closely mirrors the physical twin with complex models and data processing, which must be balanced against computational resources and available storage.

The DTRAN framework is structured across three interconnected spaces to manage complex network data [19]. Those layers are mentioned below and shown in Fig 8.

- **Physical Space**: Continuously updates the high-fidelity network infrastructure by real time data streams from diverse sources including mobile network infrastructure and satellites.
- **Cyber Space**: Utilizes graph-based data modeling and knowledge graphs to manage relationships across multi-

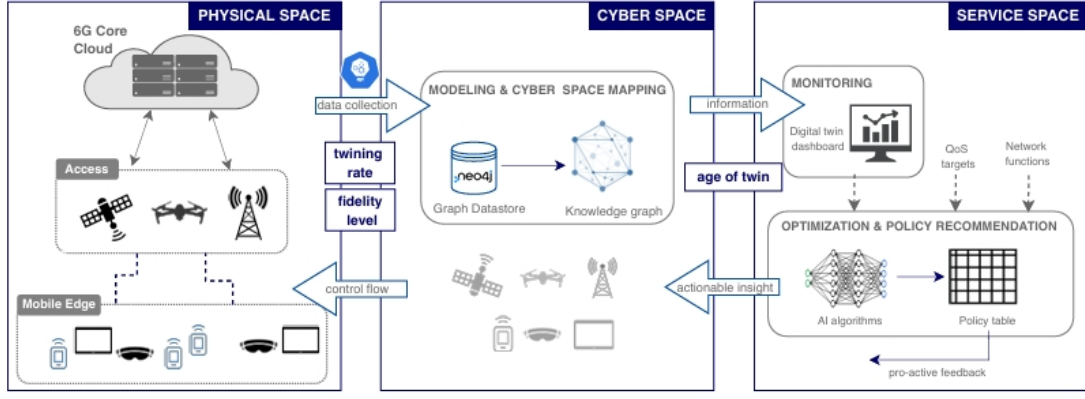


Fig. 8. Proposed DTRAN Framework [19]

sourced data

- **Service Space:** consists with enhanced Artificial Intelligence (AI) models to generate policy recommendations and optimize network functions to meet targeted Quality of Service (QoS) levels

3) *Practical use cases:* RAN configuration is combined with hundreds of parameters including tilt. Power, handover and carrier aggregation parameters. Operating across large amounts of parameters creates complexities for network operators. DTRAN model enables RAN optimization via [19]:

- **Data collection and modelling:** Using REST-ful APIs and data extraction tools, RAN data is pulled into a DT warehouse. Graph-based data modeling (via Neo4j [26]) is then applied to RAN data to preserve the relationships and increase the model accuracy via fine-tuned twining rate.
- **Monitoring and Optimization:** Basestations and User Equipments are tracked by real time models on a DT dashboard with high fidelity. Deep Reinforcement Learning (DRL) algorithms are developed to optimize configurations based on designated utility functions and constraints.
- **Application and Feedback:** The optimized configurations are tested iteratively to minimize communication failures and ensure stable age of twin values. Then, they are applied back to the RAN topology via a feedback channel.

G. Network Digital Twin for IGP Weight Optimization

NDT performs real time monitoring of the topology while interface the information from the network and apply configurations to the network automatically. Some NDTs rely on Machine Learning models, which can lead to inaccurate predictions at undesirable network conditions. It can be impacted negatively for physical network infrastructure. Existing IGP optimization methods maximize link utilization [27] [28], energy consumption [29] and delay in Service Level Agreement(SLA) but it lacks real-time autonomous optimization capabilities for the changing network conditions like traffic flows. The main benefits of using NDTs in IGP weight optimization are highlighted as follows:

- NDTs can be used to continuously monitor the network changes and optimize its IGP weights for the current traffic flows and its current topology.
- NDTs can be used to optimize the IGP metrics for the delay and loss of traffic flows directly as well.

IGP Weight Optimization primarily involves adjusting the IGP metrics of network links to optimize network performance. Open Shortest Path First (OSPF) and Intermediate System to Intermediate System (IS-IS) are two link-state routing protocols dependent on IGP weights for determining the cost of links. The IGP weight adjustments based on network monitoring can be done automatically, reducing human intervention and improving network adaptability by using NDTs.

A Network Digital Twin (NDT) represents a virtual model of a communication network, allowing testing of different scenarios without impacting the physical network. The virtual model facilitates testing various traffic flow and network configuration scenarios without affecting the real network. The optimizer in an NDT collects data from the physical network, explores different configurations virtually, and applies the optimized configuration back to the physical network. In this research, Network Decision Tool (NDT) utilizing RouteNet-Fermi, a Graph Neural Network (GNN), and a genetic algorithm has been proposed to optimize IGP weights. Inputs which are taken by RouteNet-Fermi are as follows:

- **various network parameters:** link bandwidths, queuing policies, number and sizes of queues at interfaces.
- **Traffic flow information:** average bandwidth demand, packet size, time distributions, and specified routes for each flow.

RouteNet-Fermi also known as RouteNet-F [11], acts as a machine learning-based network model and by utilizing input data it predicts the QoS metrics like delay, loss, and jitter for individual traffic flows. It is shown in Fig 3 diagram.

Online learning is used as the method for NDT to make sure that the IGP weights applied by our NDT do not impact the network negatively. It is a ML method that is trained using one data point at a time from a sequence of data. Online Gradient Descent (OGD), a supervised learning method is utilized to train a neural network based on measurements from

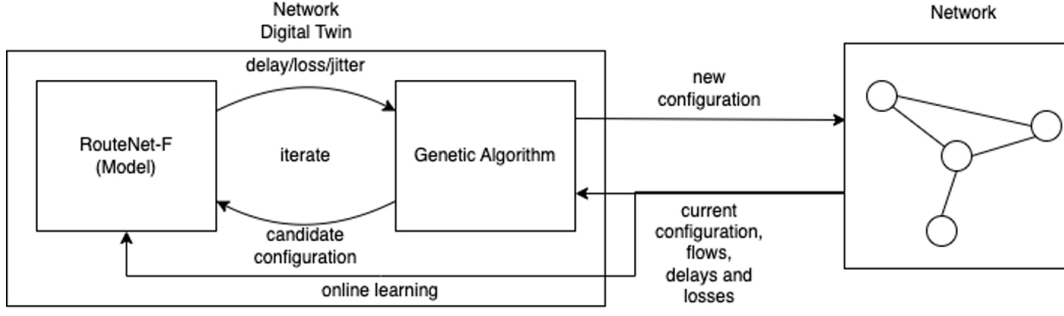


Fig. 9. Online learning NDT for Quality of Service (QoS) based IGP weight optimization [12]

the physical network. It updates neural network weights after each prediction to ensure valid IGP weight applications by the Network Delay Tool (NDT). Training RouteNet-F is triggered to achieve sufficient correlation before optimizing IGP weights when delay predictions do not align with network values. The predefined Spearman correlation threshold guides RouteNet-F training to ensure the correlation before weight optimization [12]. When the calculated correlation exceeds the threshold, network optimization takes precedence over model learning, ensuring model accuracy before IGP weight adjustments are made. A genetic algorithm is then used to explore various IGP weight configurations in RouteNet-F for improved network performance [12]. The online learning NDT for Quality of Service (QoS) based IGP weight optimization is shown in Fig. 9.

1) *Genetic Algorithm*: Genetic Algorithm is utilized for evaluate and rank each candidate IGP configuration based on a fitness function considering predicted delays and losses of traffic flows from RouteNet-F [12]. The top-ranking configuration undergoes mutation to generate the next generation. After the completion of process, new IGP weight configuration is applied to the physical network via the NDT. It is done by connecting to routers and adjusting IGP weights on interfaces. The online learning NDT updates IGP weights every 5 minutes by polling the network for its current topology and traffic flows [12]. Then it runs the genetic algorithm accordingly.

2) *Experimental Results*: The experimental evaluation of the proposed NDT demonstrates that pre-trained models can initially fail to optimize IGP weights at the high-utilization scenarios (above 70%) [12]. Preliminary test using OMNeT++ showed increased average delay of IGP weight configuration recommended by NDT. This can cause service disruptions and negatively impact the network traffic in live networks. To resolve this, a physical testbed is developed using Cisco 7200 routers and IXIA traffic generators to implement online learning framework. Spearman correlation threshold (α) is defined to NDT for determines when to prioritize fine-tuning the model's readout layers over weight optimization. According to the obtained results, fine-tuning condition can be achieved at optimal threshold (specifically $\alpha = 0.01$). Table ?? shows the impact of network delay with different α threshold values at online learning NDT. In this case, it significantly reduced average delays from 4ms to 1ms and successfully managed congestion in high-traffic scenarios [12]. Ultimately,

the study concludes that continuous online alignment between the virtual model and the physical network is necessary for maintaining accuracy and automated IGP adjustments with improved performance. Table III represents the performance variation before and after the NDT deployment for IGP weight optimization.

TABLE II
IMPACT OF α THRESHOLD ON MAX DELAY REDUCTION AND REQUIRED SAMPLES

α	Max Delay Reduction (%)	Number of Samples Required to Meet Optimization Criteria
$\alpha = 1$	4.5%	0
$\alpha = 0.03$	15%	1
$\alpha = 0.02$	14%	12
$\alpha = 0.01$	30%	25

TABLE III
NETWORK PERFORMANCE COMPARISON: PRE VS. POST NDT OPTIMIZATION

Metric	Initial State	Optimized State
Average Delay	4 ms	1 ms
Traffic Load	80 Mbps	80 Mbps
Network Status	Congested	Balanced
Data Collection Time	–	4 Minutes
Execution (Weight Change)	–	2 Minutes

H. Digital Twin Enabled Open RAN

The Open Radio Access Network (O-RAN) is a modern paradigm that transforms monolithic mobile networks through novel standards which are driven by the principles of disaggregation, cloudification, and the adoption of open interfaces. The fundamental O-RAN architecture consists disaggregated components unlike the traditional network's black-box approach. Centralized Unit(CU) and Distributed Unit(DU) are two example components which are known as E2 nodes in O-RAN architecture. Integration of intelligence to the RAN is key innovation in O-RAN. The RAN Intelligent Controller (RIC) serves as the core intelligence component of the O-RAN architecture. It is categorized into two components as Near-Real Time RIC(Near-RT RIC) and Non-Real-Time RIC based on time duration difference of the control loops they manage.

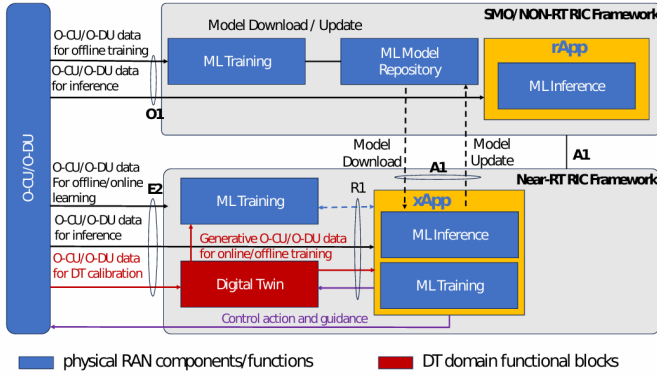


Fig. 10. DT integrated in near-RT RIC [30].

The control loop at the Near-RT RIC is ranging from 10ms to 1s duration and it can perform real-time functions at the network [23]. Microservice based applications called xApps are deployed in Near-RT RIC for perform different real-time operations like network monitoring, anomaly detecting, traffic steering and network optimization [23]. Deployment of DT for entire RAN is complex and difficult. Modular development architecture is developed for mitigate that challenge. It employs with construct the DT-RAN eventually, and design it to be open for extension as a system-of-systems [23]. In the context of Near-RT RIC, real network data is collected from E2 nodes via E2 interface as shown in 10. Those collected data is used for training and verification of AI/ML models implemented in xApps. The models in the DT is calibrated with the data captured from the E2 interface. It ensures the DT functions are behaving as closely as possible to the real network deployment for reliable RIC AI/ML model training and testing. xApps deployed in the Near-RT RIC can access DT functions via the Near-RT RIC API for AI/ML model training and testing. Simultaneous operation of multiple xApps in same network nodes at optimization tasks creates network issues. DT also be useful for detecting and mitigating conflicts among xApps. DT can predict individual and collective xApp behaviours by running them under various network conditions. It helps to understand potential conflicts and their impact on network performance while maintaining network stability. DT supports to develop coordination mechanisms and policies for xApps installing from multiple vendors. They can test their xApps on the DT platform before deployment to prevent from harmful impacts to the networks.

VI. CHALLENGES & OPPORTUNITIES

NDTs can be applied to broad variety of networking use cases with its enabled technologies and designed applications. Based on the analyzed data for DT enabled network optimization, there are few key challenges and limitations that creates future research areas.

A. Data collection and storage

In networking, collecting data and processing data is challenging and expensive task. That is because, it requires costly

telemetry systems to gather relevant network state data. Also, the data become valuable only if it has a common data format or labeling. But in real-world networks, data comes from different sources with various formats. It is important to define standard formats and interfaces that can be applicable to different sources [20]. Gathered network related data need to be stored and it requires large amount of storage. It creates another challenge to maintain large data storage to store large amount per flow records in modern networks. As a solution, monitoring platforms often use general-purpose compression methods to reduce the storage requirement but these methods do not exploit characteristics of traffic traces. It results in poor compression ratios. This invites new researches of the design of special purpose compression methods that can decrease the size of network monitoring data [20].

B. Generalization and scalability to real networks

NDTs must be capable of perform well on various network scenarios than those performed under training. Generalization of NDT models is important because training a NDT is not rapid but the network changes can happen fast [20]. As a result, it is not possible to finish the training process before initiation a new network event. Well-known ML techniques such as regularization or dropout can be applied to improve generalization of NDTs [20]. However, theses methods degrades performance especially when the network condition change drastically. Also those methods can introduce bias to the model. It creates opportunities to develop new ML models that may lead to more generalization [20]. Consequently, building scalable NDTs is an open issue that should be addressed to achieve production-ready solutions.

C. Fine-grained control and management

Low granularity is essential for perform efficient network operation. Modern communication networks carries large number of simultaneous data flows which may create scalability issues for ML-based methods. Some networking systems use traffic sampling or aggregation techniques to mitigate scalability issue. Therefore, building flow-based NDT models that can operate at a flow granularity with short time scales is an open challenge for operators and researchers.

D. Dealing with uncertainty

Neural network based models are usually deployed as black box in network systems. It is difficult to evaluate the process of certain model about the predictions made by it. It makes limitations in critical network infrastructure that are significant factors to operators for maintain robust and reliable real-world network without compromising their normal behavior. A comprehensive testing procedures are designing at controlled network environments to systematically verify that the operational conditions of DL models are within safe range before deployment on the actual network [20].

E. Model Construction

Accurately reflecting the complex mechanisms of modern networks remains a significant challenge for data-driven models. congestion control algorithms, queueing policies significantly impact on network performance and relies on end-host behavior. Additionally, routing schemes manage flows at the path level, significantly impacting overall traffic distribution. Time sensitive nature of these mechanisms makes achieving accuracy in global and local modeling exceptionally difficult [10]. Constructing data-driven models from scratch is highly resource-intensive, requiring substantial investments in data collection, model training and hyperparameter tuning [10]. To address those modeling challenges researches are introduced modular or layered conceptualizations combined with hybrid analytical-neural network architectures. Integrating analytical methods in domain knowledge can further enhance the accuracy and efficiency of these models. Transition from separately developed data-driven models to unified foundation models is one of main opportunity in this domain. Transformer architecture in natural language processing provides universal, baseline modeling capabilities for network performance evaluation which is requiring only minimal fine-tuning for specific optimization tasks [10].

VII. FUTURE DIRECTIONS

Future directions for Digital Twin (DT) enabled networks focus on overcoming current limitations in AI model generalization, data granularity, real-time synchronization and security to achieve fully autonomous, self-optimizing NGN infrastructures.

The existing AI algorithms can often fail to be scaled to larger networks or unseen situations. Further study into the field should be on enhancing model generalization, in particular, with Graph Neural Networks that can work with larger real-world networks and techniques like graph clustering to make them computationally simpler [1].

Digital Twin performance requires detailed network data to be available. The application of the Generative AI to generate high-resolution synthetic data and the creation of models that can be run at extremely fine time scales should be explored in the future to serve advanced 6G application [17]s.

It is critical to keep physical network up-to-date in line with the digital twin. Studies are required to come up with effective update systems and the two environments exchange information in a continuous and reliable manner [13].

Adaptive optimization methods are needed in future networks like Non-Terrestrial Networks, Internet of Vehicles. The future research should be on mobility prediction, efficient association of edge servers, and simultaneous optimization of various network slices [18].

With digital twins having increased control of networks, there is a need to have strong security mechanisms. In the future, it should involve more complicated cryptography [16], data integrity systems on the basis of blockchain, and explainable AI models that ensure that one can trust automated decisions [12].

VIII. CONCLUSION

Digital Twin Networks (DTNs) are becoming a major technology in the management of complex 5G and future 6G networks, and they are formed by establishing high-faith virtual approximations of physical networks. Such replicas allow real-time optimization, active scenario testing, and safe training of AI models without interfering with live network traffic. It has been reported that the combination of AI and DTNs can contribute to significant performance improvements of a network, both in terms of latency and resource usage. Models based on machine learning like RouteNet-based architectures are also extremely accurate in the prediction of network performance and they perform well in generalization to larger and previously unknown network topologies. Also, online learning and genetic algorithms are optimization techniques used in DTN frameworks, which enable automated adaptations which positively impact the efficiency of the network. Another capability of DTNs is that they enable training of Deep Reinforcement Learning agents in a safe setting by extensive trial and error using a simulated network, before deploying the optimized settings to the actual network. Although there are these strengths, more investigation will be needed before fully autonomous self-optimizing networks can be attained through refinements to the AI model to make it manage great amounts of data, real-time synchronization between the digital and physical network, and the introduction of robust security measures.

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